

You can (and should!) use statistical software to solve these problems. However, your work must be documented. Please send your solutions along with any code, plots, etc. used. In each problem, be sure to state which statistical tests you have completed, whether or not all of the assumptions of the test you used are satisfied, and what confidence level you have used. Please summarize in complete sentences.

- (1) Two datasets are included on the course page: winequality-red, and winequality-white. These two datasets are related to red and white variants of the Portuguese “Vinho Verde” wine. Your goal is to determine whether the data suggest that there is a difference in pH between the red and white varieties.

- (2) The Boston Housing dataset contains information collected by the U.S Census Service concerning housing in the area of Boston Mass. It has been used extensively throughout the literature to train ML algorithms. The data was originally published by Harrison, D. and Rubinfeld, D.L. ‘Hedonic prices and the demand for clean air’, J. Environ. Economics & Management, vol.5, 81-102, 1978. More information about it is given here <https://www.cs.toronto.edu/~delve/data/boston/bostonDetail.html>.
 - (a) We will consider the variable NOX and see its correlation with other variables in the dataset. What is the variable NOX measuring? What other variables in the dataset do you think it is correlated with (answer this before any statistical analysis)? Explain why you think so.
 - (b) Create the following scatterplots: AGE vs NOX, DIST vs NOX and log(DIST) vs NOX. (For the last one I want you to take the log of each DIST, you can do this by simply writing log(DIST)). Comment on the features of the scatterplots. Do these variables seem linearly related? Does a linear relationship make sense intuitively? Do you think that pollution (NOX levels) decreases exponentially as you move away from a city center?
 - (c) Fit a linear regression model to each of the above variables ($\text{NOX} \sim \text{AGE}$, $\text{NOX} \sim \text{DIST}$, and $\text{NOX} \sim \log(\text{DIST})$). Be sure to include the regression coefficients. Comment on the correlation of these variables.
 - (d) Provide residual plots for each of your regression models. What do these plots tell you about the linear models you fit?

- (3) Consider the R data set ‘ChickWeight’ (also available on the course page). This data consists of the body weights of the chicks measured at birth and every second day thereafter until day 20. They were also measured on day 21. There were four groups on chicks on different protein diets labeled 1,2,3,4. On day 10, does the data suggest that the protein diet has an effect on the chick weights? Does this still occur on day 21?